Tactix interim N4 audit: current implementation versus ledger

Codex for Andrew

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Outcome. Three high-severity scheduling-parity rows incorrectly said their mechanisms were absent from production. Current source, Git history, and focused current-source Debug tests show they shipped in commit `7a668e76`: Congruence's redundant-prefix cleanup, Reduce's newly derived-root guidance, and Vivify's learned-large BVE retry marks. The living ledger and implementation note now distinguish *implemented* from *accepted*. No solver code or schedule changed. The exact source/test mapping is in the AM440 technical report under `benchmarks/reports/`.

Validation and limitation. Current-source Debug tests `congruence`, `reduce_root_guidance`, `vivify`, and `vivify_binary_retry` have passed within job 1723391. The job had passed 186 of 187 tests and was still running its final scripted demo at this snapshot. Its partial status is not a full Debug gate. The historical Congruence and Reduce runtime experiments remain useful evidence but do not establish current-source cumulative Rank/B28+C5 or N4 acceptance. Required affected fuzz, full-trajectory comparison, and runtime attribution remain open for these mechanisms.

Audit-tool repair. The configuration-wiring check treated `sweep_incomplete`, a Sweep-owned resumable-pass state latch, as an unclassified strategy setting. It is now explicitly excluded as lifecycle state, rather than given a fake policy writer. The check passes with 96 actual configuration fields and zero failures; the option-coverage check accounts for all 418 pinned reference options. Existing documentation-size warnings remain; no compaction was performed.

Next actions. Finish the running full Debug and paired AES certificate checks, resolve H2's general proof/checker-state mismatch, complete the N1–N4 and winner-rule gates, then prepare the final learning readiness report. The requested full 1,166-instance Tier D comparison at 5,000 seconds, 4 CPUs and 32 GiB on hwulab remains deferred until the preceding goal is verified.